

Within limits, above the guideline

Five years of annual PM2.5 readings checked against EU and WHO thresholds

THE CLAIM UNDER REVIEW

***All available PM2.5 annual means were
above WHO guidance and below the EU limit***

CDE reproduces Newsbook's PQ comparison · 19 July 2025

The Minister tabled the figures; the compliance statement is our test of that table.

VERDICT: LARGELY SUPPORTED

All annex values are under 25; one EEA check differs

TL;DR

A CDE briefing dated 19 July 2025 reproduces Newsbook’s summary that five Maltese monitoring stations were above the World Health Organization’s PM2.5 guideline but within the EU limit. The source behind the comparison is Parliamentary Question 29696: the Minister tabled annual station values for 2020–2024. The answer itself is procedural; the numbers, not a ministerial statement of compliance, are what this check tests.

- 1

The historical EU comparison holds for reported values. All 23 values supplied are below the 25 µg/m³ annual EU limit applicable during 2020–2024.
- 2

Every reported value is above WHO’s health-based guideline. The annual means range from 7.4 to 13.9 µg/m³; the WHO guideline is 5 µg/m³.
- 3

Two station-years are missing. St Paul’s Bay is marked n/a in 2020 and 2021, so the table cannot establish compliance at all five stations for every year.
- 4

The next EU standard is stricter. Fourteen of 23 values exceed the 10 µg/m³ annual limit the EU requires states to attain by 2030. This is context, not a breach during 2020–2024.
- 5

An independent EEA cross-check is mostly consistent. The EEA validated daily files cover 11 station-years in 2020–2024. Ten match the annex when rounded to one decimal; Attard 2024 does not.
- 6

Verdict: largely supported (moderate confidence). The comparison holds for all 23 annex values. Two entries are missing, and one current EEA cross-check differs from the annex.

THIS CLAIM

SUPPORTED

LARGELY SUPPORTED

NOT SUBSTANTIATED

MISLEADING

CONTRADICTED

23/23

reported values above WHO’s 5 µg/m³ guideline

23/23

reported values below the then-applicable EU 25 µg/m³ limit

14/23

above the EU 10 µg/m³ limit due by 2030

WHAT WOULD MOVE THE VERDICT UP

ERA’s validated annual series with completeness notes, and an explanation for the Attard 2024 difference.

WHAT WOULD MOVE IT DOWN

A corrected official table showing an annual mean above 25 µg/m³, or evidence the table’s values are not comparable to that limit.

IN THIS REPORT	
1 What was asked and answered	4 Interpretation and limitations
2 Standards and method	5 Verdict and evidence needed
3 What the table shows	6 Sources

1 What was asked and answered

On 16 July 2025, MP Rebekah Borg asked the Minister for the Environment, Energy and Public Cleanliness for the annual PM2.5 means recorded by every ERA monitoring station for the previous five years. In her answer, Minister Miriam Dalli said the requested information was being laid on the Table of the House. The 1-page annex provided the station-by-station values reproduced in this report [1].

THE OFFICIAL ANSWER

“Ninforma lill-Onor. Interpellanta li qed inpoġġi l-informazzjoni mitluba fuq il-Mejda tal-Kamra.”

Translation: “I inform the Honourable Interpellant that I am laying the requested information on the Table of the House.” This does not state that the values comply with an EU standard. The compliance statement is an interpretation of the table.

The proposition assessed is Newsbook’s reported comparison, reproduced by CDE, that the recorded station means were above WHO’s guideline and within the EU limit. It is not presented as a direct quotation from the Minister. The parliamentary portal lists the question as Legislature XIV, Sitting 368, question 29696, 16 July 2025 [1].

Year	Attard	Msida	St Paul’s Bay	Żejtun	Għarb
2020	9.8	11.0	n/a	9.6	7.4
2021	11.9	12.4	n/a	10.7	8.6
2022	10.8	11.7	10.6	10.1	8.9
2023	11.4	13.9	9.5	10.9	8.9
2024	11.9	11.6	9.5	10.4	8.7

Table 1. Annual PM2.5 concentration in µg/m³, transcribed from the annex to PQ 29696 [1]. “n/a” is retained as printed.

2 Standards and method

We compared every value in the parliamentary annex with three different reference points. The annual EU limit applicable during 2020–2024 was 25 µg/m³. The WHO’s 2021 air-quality guideline recommends 5 µg/m³ as the annual mean; it is health-based guidance, not the EU legal limit [2–3]. Recast EU Directive 2024/2881 sets a 10 µg/m³ annual limit to be attained by 1 January 2030 [3]. We keep that future standard separate from compliance in the years reported.

The annex has 25 possible station-year cells. We transcribed its 23 numeric values and kept the two St Paul’s Bay n/a entries missing rather than imputing them. `tools/cc-007-report/calc.py` recalculates counts, range and station means from `data/cc-007/station_pm25.csv`; the formulas and results are saved in `data/cc-007/checks.csv`.

Evidence grades: the parliamentary answer is grade C (official monitoring data reproduced in an answer); the WHO guideline is grade C (evidence-based professional guidance); the EU limit is grade C (primary legislation). This is a standards-and-record check, not an independent exposure or health assessment. The WHO guideline is informed by systematic reviews of health evidence [2, 4].

3 What the table shows

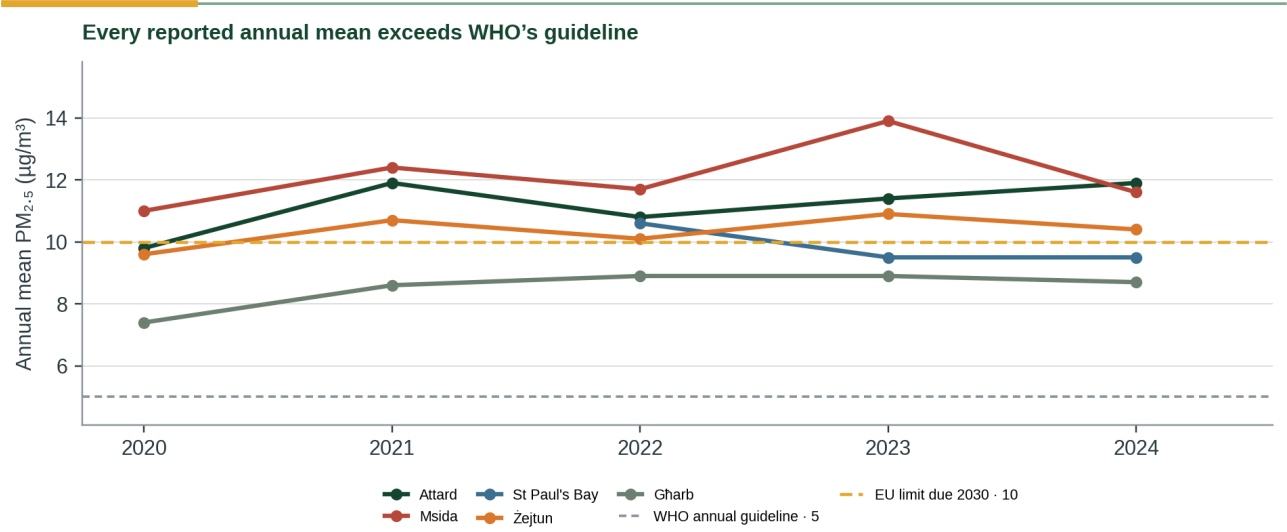


Figure 1. Annual means at five ERA stations. Dashed lines mark the WHO guideline (5 µg/m³) and the EU standard due by 2030 (10 µg/m³). All reported values are also below the 25 µg/m³ EU limit applicable to 2020–2024. No value is plotted for St Paul's Bay in 2020 or 2021.

Measure	Result	Interpretation
Reported values	23 of 25	Two station-years unavailable: St Paul's Bay, 2020–21.
WHO annual guideline (5 µg/m³)	23/23 above	Every reported station-year exceeds the health-based guideline.
EU annual limit in 2020–2024 (25 µg/m³)	23/23 at or below	All available values meet the historical legal limit.
EU limit due by 2030 (10 µg/m³)	14/23 above	A forward-looking comparison; not the applicable limit in 2020–2024.

Across the 23 supplied observations, annual means range from 7.4 µg/m³ (Gharb, 2020) to 13.9 µg/m³ (Msida, 2023). By station, means of the available years range from 8.50 µg/m³ at Gharb to 12.12 µg/m³ at Msida. The St Paul's Bay mean uses only 2022–2024, since the first two years are absent. These are descriptive arithmetic means of the annual values in the annex, not multi-year regulatory statistics.

4 Interpretation and limitations

The three comparisons answer different questions. The 25 µg/m³ EU limit is a legal compliance benchmark for the period under review. The WHO value is a health-based guideline; being below a legal limit does not mean air pollution has no health relevance. The 10 µg/m³ EU standard signals a future regulatory direction. It is not valid to present the 2020–2024 readings above 10 as breaches of a limit that is to be attained in 2030 [2–3].

Compliance and health are distinct		SUPPORTED WITH CAVEAT
EVIDENCE FOR THE CLAIM'S IMPRESSION Every numeric value in the official table is below the EU's 25 µg/m³ limit applicable to the reported years.	EVIDENCE AGAINST Every numeric value is above the WHO 5 µg/m³ annual guideline; 14 also exceed the future EU 10 µg/m³ standard.	
WHY THEY DIFFER, AND WHAT IT MEANS HERE The legal limit, health guideline and future EU standard have different purposes and dates. The table supports historical EU compliance for the available data, but not a general claim that the air is healthy or that every station complied in every year.		

The EEA Air Quality download service publishes validated station measurements. We averaged its valid daily PM2.5 aggregates for each calendar year with available coverage from 2020 to 2024. This reproduced 11 of the annex’s 25 possible station-years: ten values round to the annex at one decimal place. The EEA mean of valid daily aggregates for Attard in 2024 is 12.122 µg/m³, while the annex reports 11.9. Both are above the WHO guideline and below the 25 µg/m³ EU annual limit; the 0.2 difference remains unexplained. The retrieved EEA files provide no validated daily series for the other 14 station-years, including St Paul’s Bay in 2020–21. Absence from those files does not establish that measurements were not made. Derived values, valid-day counts, source URLs and file hashes are saved in *data/cc-007/eea_validated_crosscheck.csv* and *data/cc-007/eea_station_file_manifest.csv*; regenerate them with *tools/cc-007-report/eea_crosscheck.py* [5].

Neither the parliamentary table nor this partial EEA check provides a full basis for local exposure or individual health outcomes. ERA’s validated annual series, annual completeness notes and an explanation of the Attard difference are still needed for a complete reconciliation.

The official annex’s two n/a cells are material to the phrase “all five stations” across the full period. They are not evidence that St Paul’s Bay exceeded the legal limit; they leave those two station-years unverified.

5

Verdict and evidence needed

VERDICT

LARGELY SUPPORTED

All reported values meet the applicable EU limit and exceed WHO’s guideline; two station-years are missing.

The reported comparison is supported for all 23 numeric observations in the annex: each is below the EU annual limit of 25 µg/m³ in force for 2020–2024, and each exceeds the WHO annual guideline of 5 µg/m³. The EEA check independently supports this broad comparison for 11 station-years, although one rounded table value differs. Confidence remains moderate because 14 station-years were unavailable in the retrieved EEA files, the annex marks two St Paul’s Bay values n/a, and Attard’s 2024 value needs reconciliation. The Minister’s answer tables figures but does not herself claim compliance.

A complete comparison needs ERA’s validated annual series with completeness and quality-control notes, the missing St Paul’s Bay results if they exist, and an explanation for the Attard 2024 difference.

This draft records the maintainers’ current instruction to handle right of reply later. No reply was requested for this draft; the status remains a draft for maintainer review.

Reproducible files: *data/cc-007/station_pm25.csv*, *data/cc-007/checks.csv*, *data/cc-007/eea_validated_crosscheck.csv* and scripts under *tools/cc-007-report/*. The exact Maltese question and answer, transcribed annex and attachment hashes are in *literature/CC-007/primary-source.md*.

6

Sources

- [1] House of Representatives (2025). Parliamentary Question 29696, Legislature XIV, Sitting 368, 16 July 2025; oral question and tabled annex. <https://pqs.parlament.mt/?pqid=68a890cbcb302292bfff56dce>
- [2] World Health Organization (2021). WHO Global Air Quality Guidelines, Table 3.26: annual PM2.5 guideline 5 µg/m³. <https://iris.who.int/bitstream/handle/10665/345329/9789240034228-eng.pdf>
- [3] European Parliament and Council (2024). Directive (EU) 2024/2881, Annex I: 25 µg/m³ limit through 11 December 2026; 10 µg/m³ to be attained by 1 January 2030. <https://eur-lex.europa.eu/eli/dir/2024/2881/oj/eng>
- [4] Pérez Velasco R., Jarosińska D. (2022). Update of the WHO global air quality guidelines: systematic reviews – An introduction. *Environment International* 170:107556. doi:10.1016/j.envint.2022.107556. WHO guideline-development context. <https://doi.org/10.1016/j.envint.2022.107556>
- [5] European Environment Agency (accessed 3 October 2026). Air Quality Download Service, verified E1a station data. Annual figures here are means of valid daily aggregates and are used as a partial cross-check. <https://air.discomap.eea.europa.eu/arcgis/rest/services/AirQuality/AirQualityDownloadServiceEUMonitoringStations/MapServer/0>
- [6] CDE (published 19 July 2025). Malta News Briefing – Saturday 19 July 2023. Reproduces the Newsbook PM2.5 comparison and attributes it to Newsbook; secondary source only. <https://cde.news/malta-news-briefing-saturday-19-july-2023/>
- [7] Newsbook (19 July 2025). “Air pollution exceeds WHO guidelines throughout Maltese islands.” Original secondary report; a Wayback snapshot is listed in the archive manifest. <https://newsbook.com.mt/en/air-pollution-exceeds-who-guidelines-throughout-maltese-islands/>

Appendix A: series standards and verdict scale

Every investigation in this series follows the same rules so that verdicts are comparable and challengeable.

Verdict	Meaning
SUPPORTED	Evidence consistently backs the claim as stated.
LARGELY SUPPORTED	Backed by the evidence, with minor caveats that do not change the substance.
NOT SUBSTANTIATED	Stated more strongly than the evidence offered or available allows. May be true in some conditions but has not been shown.
MISLEADING	Omits material facts so that the overall impression is inaccurate, even if individual statements are defensible.
CONTRADICTED	The available evidence points against the claim.

Confidence

High: multiple independent lines of evidence agree. **Moderate:** evidence is relevant but incomplete or indirect. **Low:** evidence is thin, second-hand or conflicting.

Evidence grades

A experiment · B observational study with a control or gradient · C review, professional guidance, official statistics or primary legislation · D assertion or anecdote. The official parliamentary table is the direct evidence for the recorded annual means; this report does not infer exposures or health outcomes beyond those readings.

Standards

- **Quote the speaker, not the headline**, with outlet and date, and link the original where possible.
- **Separate the claim from the evidence offered for it**, and say who made which statement.
- **Grade the evidence** and show differing views side by side where the evidence is unsettled.
- **State what we could not access or verify**, and what would change the verdict either way.
- **Ask for the missing evidence** in specific, answerable requests.
- **Offer a right of reply** before wider circulation and append responses.
- **Correct errors openly** with a version number and change log.
- **Whistle-blow only on evidence:** a verdict of *Misleading* or *Contradicted* requires documents or data that can be shown, not inference.
- **Check every side.** Government, opposition, NGOs and developers are held to the same standard.

Appendix B: revision log

Version	Date	Change
1.0	2 Oct 2026	First draft based on Parliamentary Question 29696 and its tabled annex.
1.1	3 Oct 2026	Added an independent partial cross-check against EEA validated station data; documented one unresolved Attard 2024 difference.